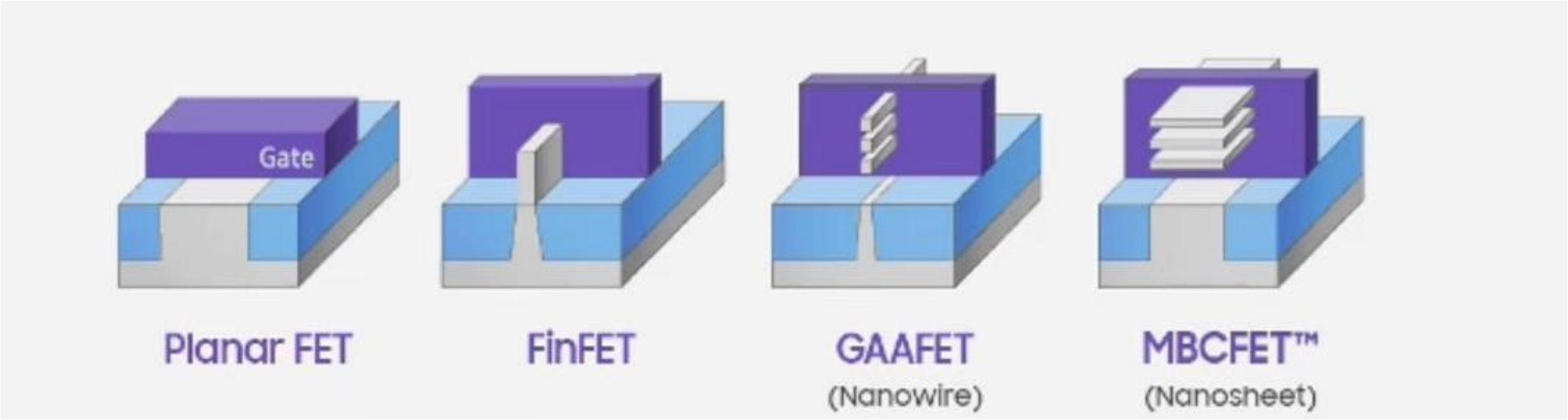


Fig 1. Evolution of transistor gate design [1]

The Evolution of Transistor architecture

By Nathan LeBlanc

Why scaling stopped working

Moore's Law prediction of a two-year doubling of transistor density has slowed throughout the 21st century due to three major limiting factors.

1. Physical scaling limits (quantum tunneling, Short Channel Effects)
2. Economic Constraints (rising R&D and lithography costs)
3. Thermal management (increasing power density limiting heat dissipation

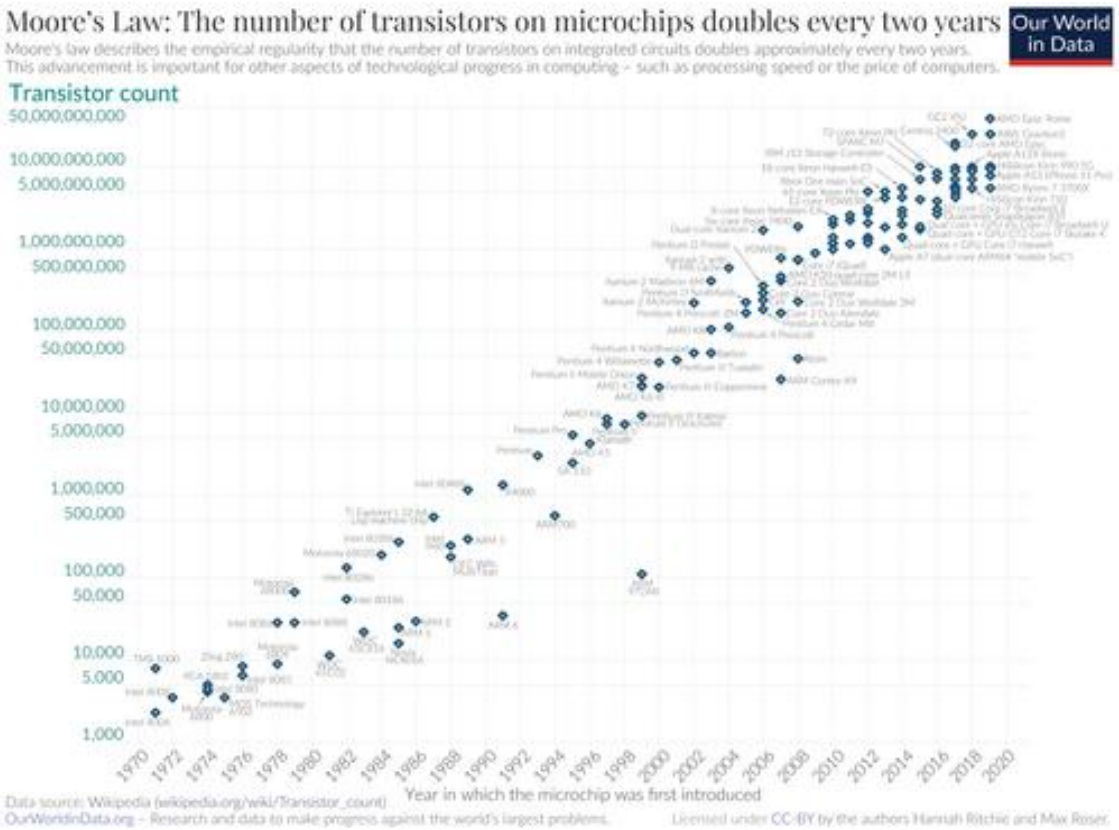


Fig 2. Moore's law transistor count over years [2]

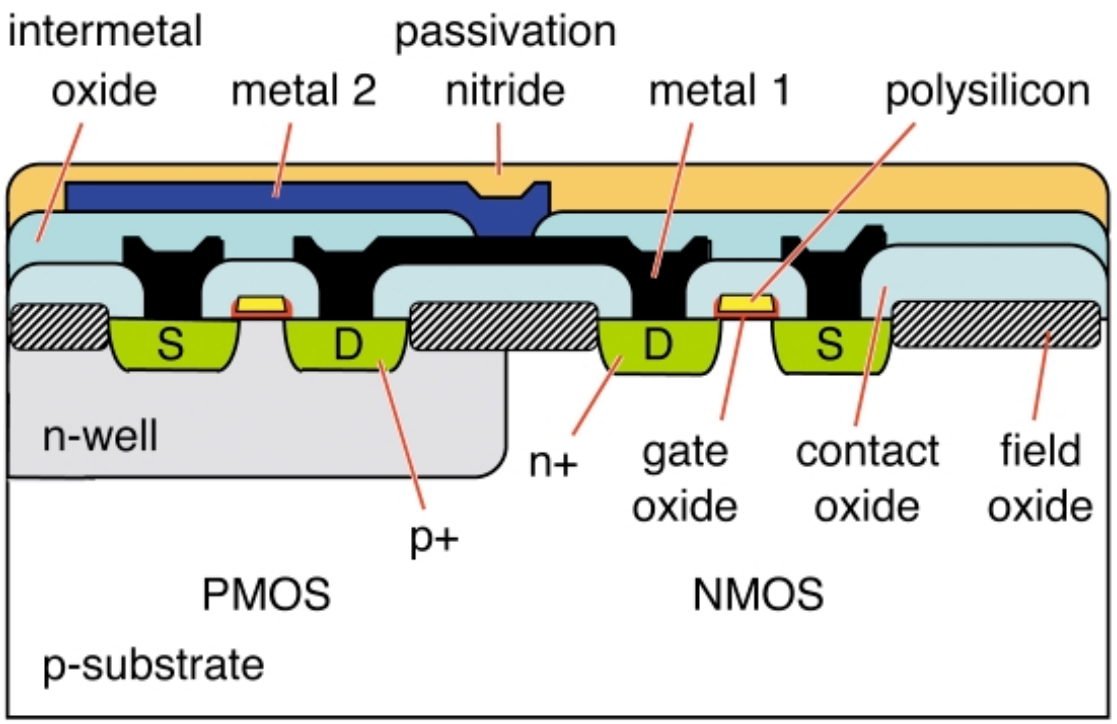


Fig 3. Planar CMOS cross section [3]

Planar CMOS core steps

- Substrate preparation (cleaning)
- Active region definition and isolation (oxidation, protective material deposition, STI)
- Well formation (ion implantation and annealing)
- Gate dielectric and polysilicon deposition
- Gate patterning
- Source/drain formation (ion implantation)
- Dielectric deposition and contact formation
- Metallization
- Passivation

Why planar failed and FinFET was needed

When the length of the planar MOSFET gate began to be reduced below about 40 nm, the influence of short channel effects, threshold voltage roll off and drain induced barrier lowering made it difficult for a single gate surface to provide adequate electrostatic control of the channel. This was resolved with the use of a FinFET that wrapped a single gate surface around all three sides of a vertically oriented silicon fin, which restored the ability to continue reducing the size of devices well beyond 22 nm

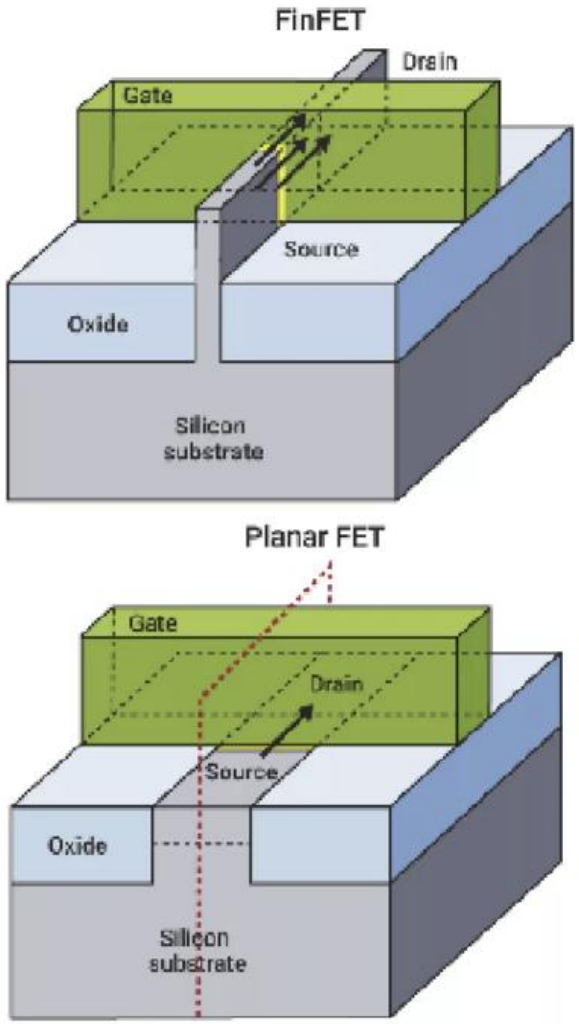


Fig 4. 3D image of planar MOSFET vs FinFET [4]

Self-Aligned (Double/quadruple) patterning

193 nm immersion lithography cannot resolve features below 45 nm half-pitch. single exposure was insufficient

Self-alignment is inherent with both SADP and SAQP, both patterns produced in one process cycle, eliminating overlay error of LELE

SAQP repeats the cycle of SADP achieving 19 nm pitch enabling 5-7 nm wide fins at 7 nm and 5 nm nodes.

Yield drivers: spacer width non-uniformity, LER, and CDU all translate directly to fin width variation and threshold voltage shift across the die

1. Mandrel formation
2. Spacer deposition
3. Spacer formation
4. Mandrel removal
5. Etch back
6. Final pattern

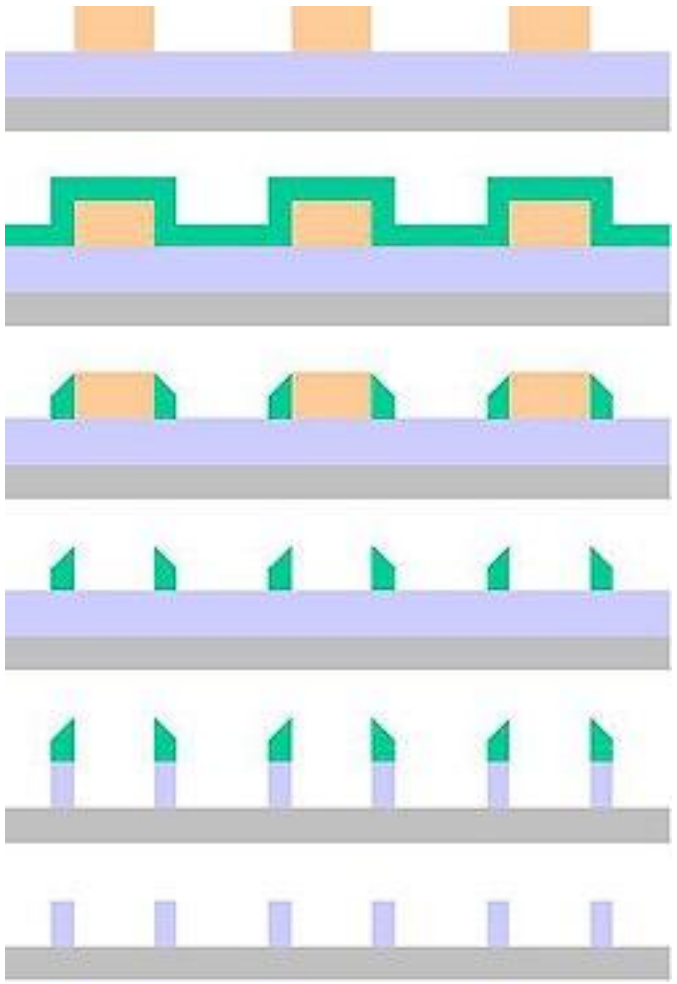


Fig 5. SADP fabrication diagram [5]

High-k/Metal Gate

- When SiO2 gate dielectric scaled below 2nm, quantum mechanical tunneling caused unacceptable gate leakage current and device degradation
- Polysilicon gate caused poly depletion this effectively increased EOT, degraded drive current and threshold voltage control
- HfO2 replaces SiO2 with a dielectric constant of 20–25 vs 3.9, allows physically thicker film at same EOT eliminating tunneling
- TiN and TaN metal gates replace polysilicon which eliminates poly depletion, enables precise work function tuning
- ALD is essential for FinFET HKMG as sputtering cannot conformally coat three-sided fin geometry, ALD provides angstrom-level uniform coverage

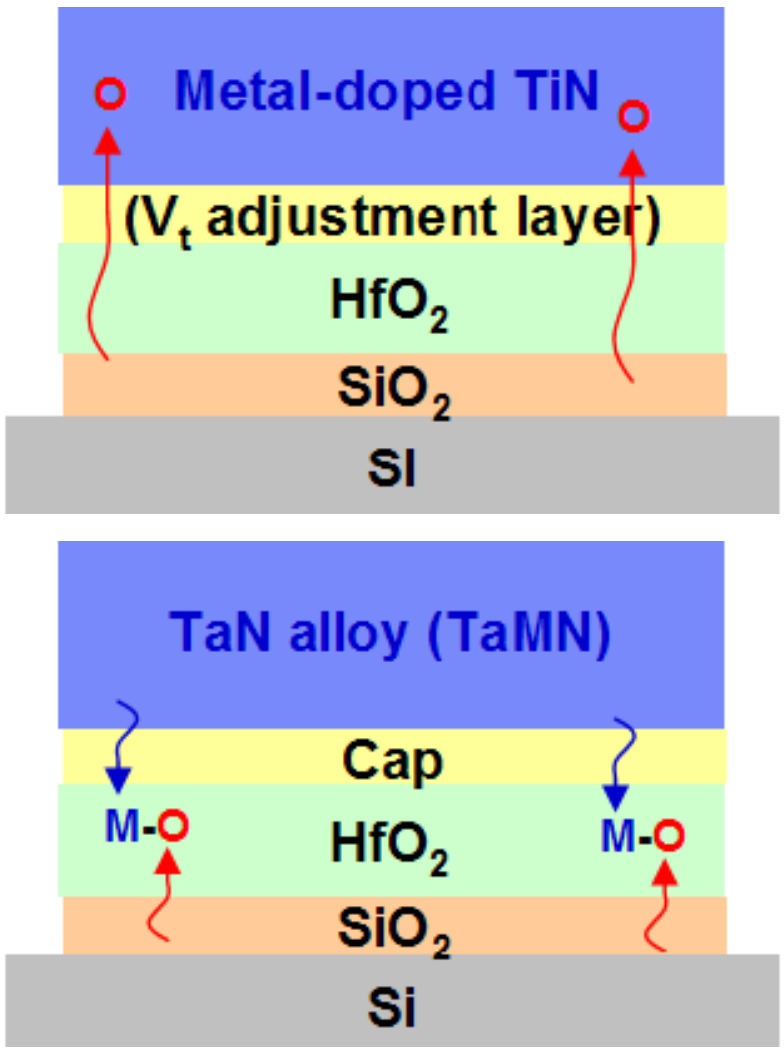


Fig 6. high-k/metal gate stack using TiN and TaN [6]

Why FinFET failed and GAAFET was needed

- Below 7 nm fin widths become too narrow to maintain structural integrity and process control.
- Tri-gate FinFET loses electrostatic control as gate length scales and short channel effects return
- Gate-All-Around wraps the gate on all four sides of stacked silicon nano sheet/wire, maximum electrostatic control at sub-5 nm nodes

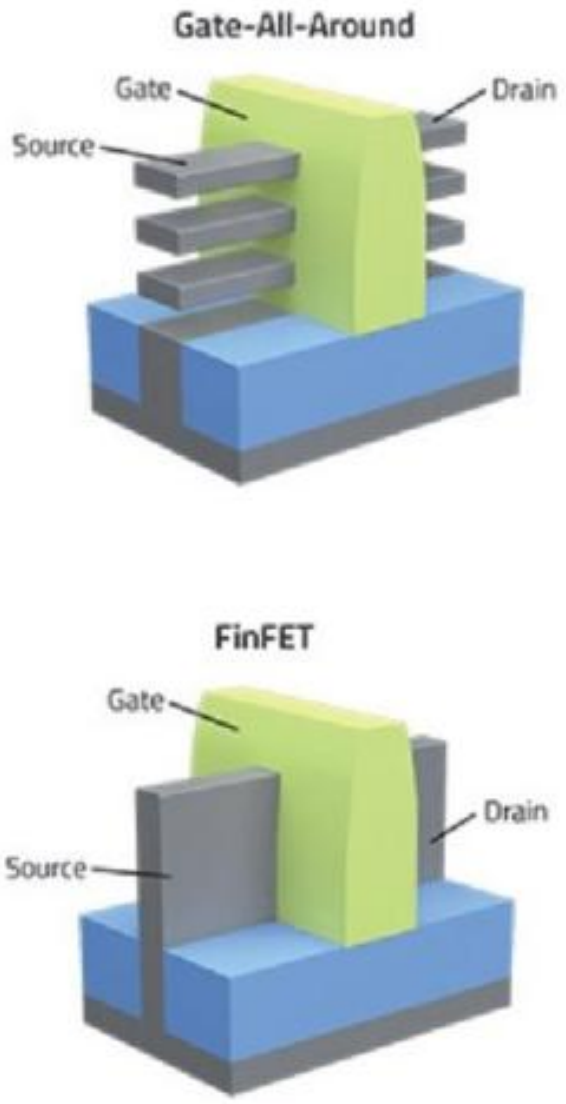


Fig 7. 3D image of GAAFET vs FinFET [7]

Epitaxial Si/SiGe Superlattice Stack Formation

- Replaces bulk silicon substrate as channel definition, and is the first fundamental departure from all prior FET fabrication
- Grown by Reduced Pressure CVD (RPCVD) below 650°C using DCS and GeH4 as precursors
- Alternating Si (channel) and SiGe (sacrificial) layers at 8-10 nm each
- Ge concentration 20-30 atomic percent. optimized for etch selectivity during channel release without inducing dislocation defects
- Layer thickness uniformity and Ge concentration are critical, any variation degrades electrostatic control and reduces yield

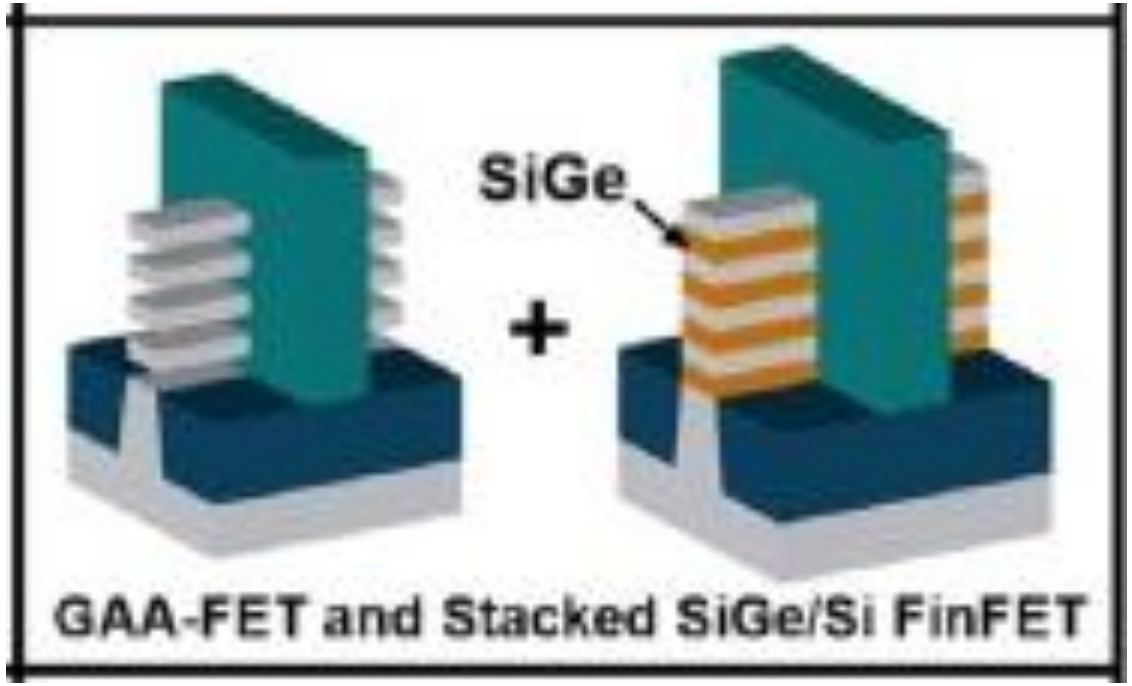


Fig 8. image of Epitaxial Si/SiGe superlattice stack [8]

Inner Spacer Formation

- Purpose: isolate gate from source/drain between nanosheets and reduce parasitic gate-to-S/D capacitance
- For Lateral isotropic SiGe cavity etch gas phase etching preferred, SiGe/Si selectivity ~9x greater than ICP
- Yield and reliability: moon-shaped spacer profile risks TDDB thus more recent process development targets square profile for improved uniformity

a. Starting Si/SiGe superlattice stack
b. Selective lateral recess etch
c. Inner spacer deposition
d. Spacer etch back

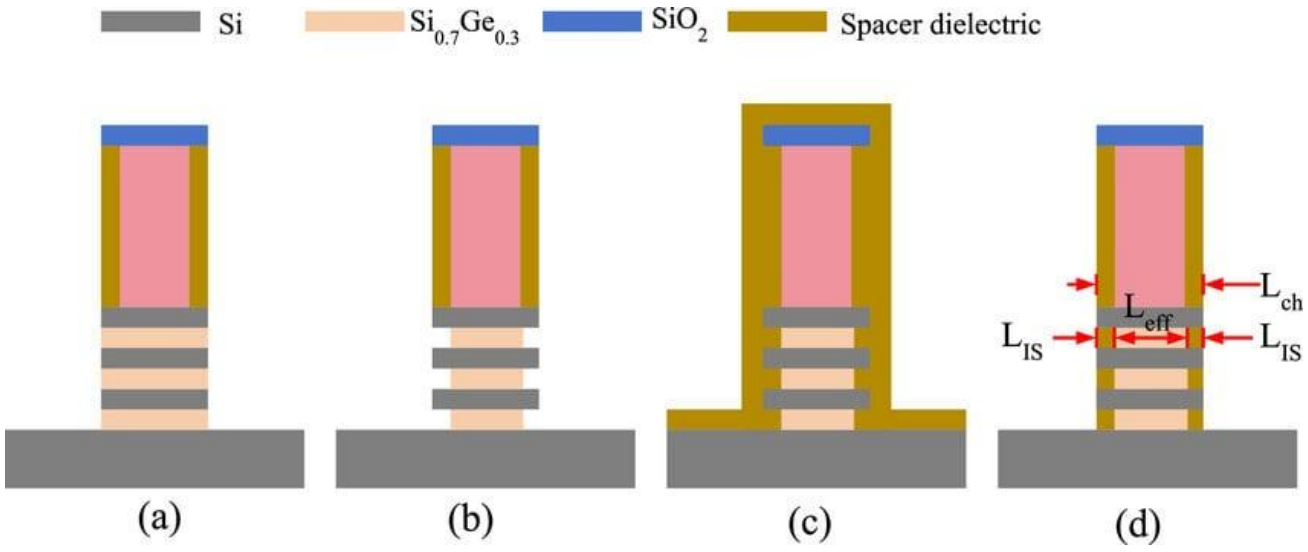


Fig 9. cross section image of inner spacer formation [9]

Conformal Atomic Layer Deposition for Gate Stack Deposition

- Suspended nanosheets exposed on all four sides, PVD sputtering physically impossible in enclosed sub-10 nm gaps
- ALD self-limiting surface reactions, one monolayer per cycle, 0.5–1.5 Å per cycle, geometry independent
- Gate stack: SiO2 interfacial layer → HfO2 high-k → TiN work function metal → W fill metal
- Cumulative stack thickness must fit within 8–10 nm inter-sheet gap, no room for non-uniformity
- Yield challenge: top-to-bottom diffusion gradient causes thicker gate dielectric on upper sheets than the lower, introduces intra-device threshold voltage variation unique to GAAFET

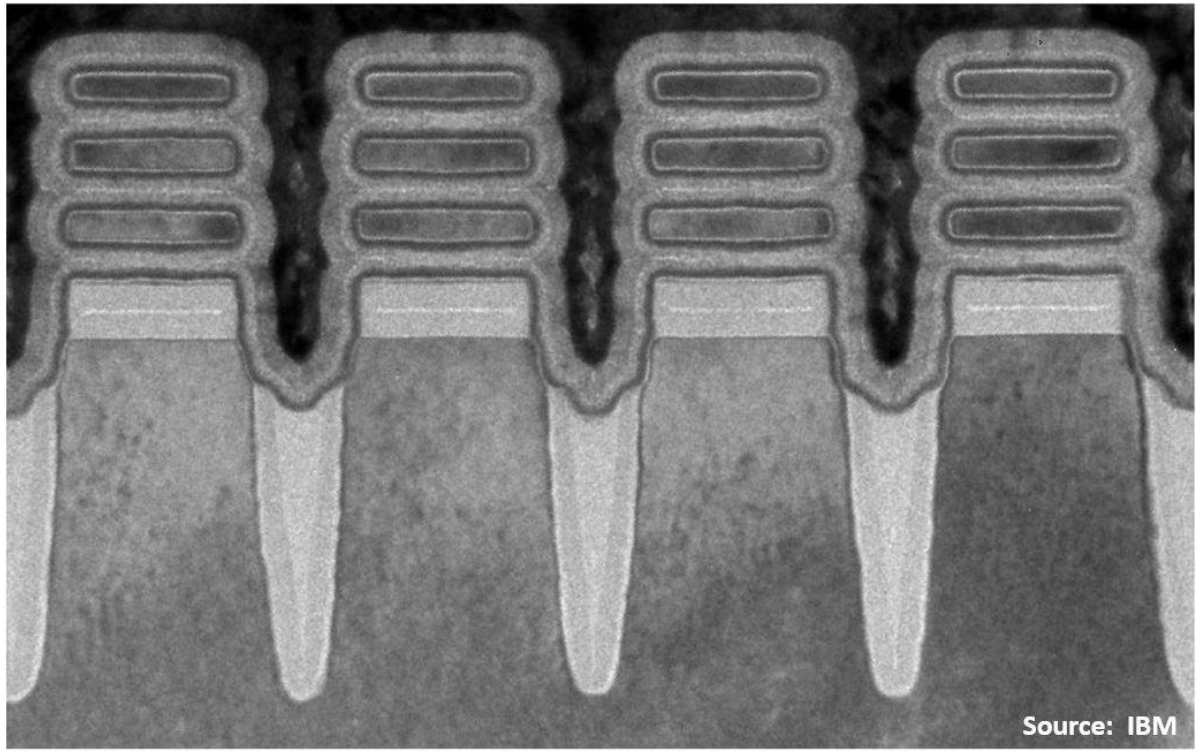
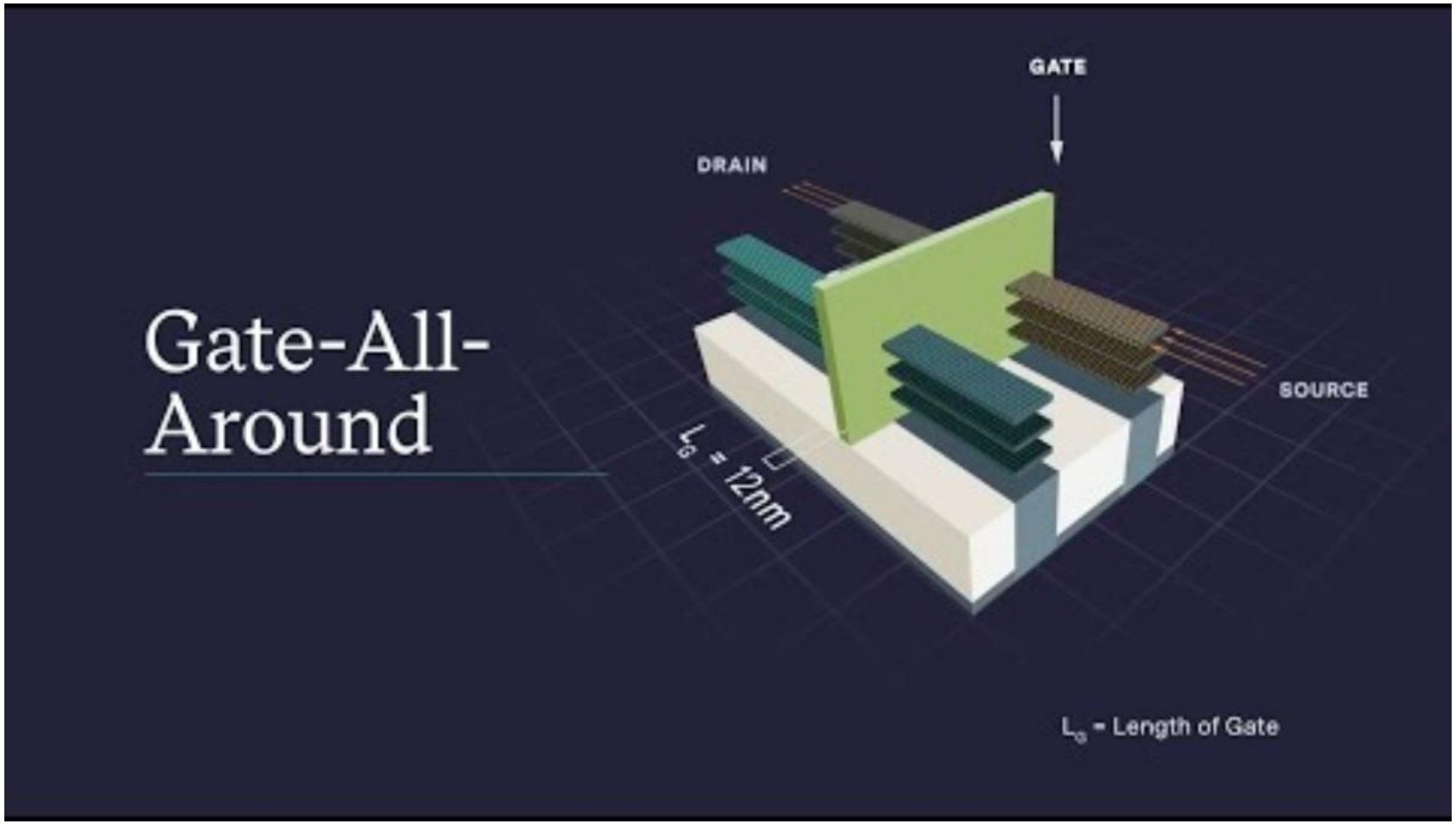


Fig 10. TEM cross image of NanoSheet GAAFET [10]

Conclusion

- Planar CMOS established the foundational process vocabulary. isolation, gate oxidation, self-aligned patterning, doping, and metallization. sustaining scaling from the micron scale to approximately 40 nm
- FinFET extended scaling through SADP/SAQP multi-patterning and HKMG with conformal ALD, replacing planar geometry with a three-sided gate on a vertical fin to restore electrostatic control
- GAAFET pushed further through epitaxial superlattice growth, selective channel release, inner spacer formation, and conformal ALD gate deposition, this enabled full four-sided gate enclosure of stacked silicon nanosheets at sub-5 nm nodese-sided gate on a vertical fin to restore electrostatic control
- Each architectural transition was not a design choice; it was a fabrication-driven response to the physical limits of the generation before it.

What’s next



https://youtu.be/ST_1krgYni4?si=wg1ivZkkbdNvIGbG

Bibliography

[1] "GAA-FET diagram," *SemiWiki*, Sept. 2025. [Online]. Available: https://semiwiki.com/wp-content/uploads/2025/09/GAA-FET-min.jpg. [Accessed: April 4, 2026].

[2] M. Roser, "Moore's Law: The number of transistors on microchips," *Wikimedia Commons*, 2020. [Online]. Available: https://upload.wikimedia.org/wikipedia/commons/0/00/Moore%27s_Law_Transistor_Count_1970-2020.png. [Accessed: April 4, 2026].

[3] A. Hierlemann and H. Baltes, "CMOS-based chemical microsensors," *Sensors and Actuators B: Chemical*, vol. 93, no. 1-3, pp. 29-48, 2003. [Online]. Available: https://www.semanticscholar.org/paper/CMOS-based-chemical-microsensors.-Hierlemann-Baltes/0add9c5b31d1cc089231c962ca270da017e3b25a. [Accessed: April 4, 2026].

[4] "Structure of Planar FET versus FinFET," *ResearchGate*, Dec. 2023. [Online]. Available: https://www.researchgate.net/figure/Structure-of-Planar-FET-versus-FinFET-The-FinFET-Fin-Field-Effect-Transistor-as-is_fig1_376672940. [Accessed: April 4, 2026].

[5] "Spacer Patterning process diagram," *Wikimedia Commons*, [n.d.]. [Online]. Available: https://upload.wikimedia.org/wikipedia/commons/1/18/Spacer_Patterning.JPG. [Accessed: April 4, 2026].

[6] D. J. Frank, "High-k/metal-gate innovations enabling continued CMOS scaling," *Proceedings of the IEEE*, vol. 99, no. 2, pp. 258-269, Feb. 2011. [Online]. Available: https://www.semanticscholar.org/paper/13c36d88c28b2a78cc955c9119d06b88aa312b4c. [Accessed: April 4, 2026].

[7] "Comparative structures of MOSFET, FinFET and GAAFET," *ResearchGate*, March 2024. [Online]. Available: https://www.researchgate.net/figure/Comparative-structures-of-MOSFET-FinFET-and-GAAFET_fig3_379259370. [Accessed: April 4, 2026].

[8] "Evolution of transistor architectures," *Elsevier / Microelectronic Engineering*, 2023. [Online]. Available: https://ars.els-cdn.com/content/image/1-s2.0-S0167931723000588-gr1.jpg. [Accessed: April 4, 2026].

[9] "Process of inner spacer module," *ResearchGate*, July 2024. [Online]. Available: https://www.researchgate.net/publication/382538027/figure/fig1/AS:1143128173105707 0@1763470936719/Process-of-inner-spacer-modulea-source-drain-recess-to-expose-the-inner-spacer.jpg. [Accessed: April 4, 2026].

[10] "Row of 2 nm nanosheet devices," *Semiconductor Digest*, May 2021. [Online]. Available: https://www.semiconductor-digest.com/wp-content/uploads/2021/05/Fig-1_Row_of_2_nm_nanosheet_devices-c-ann.jpg. [Accessed: April 4, 2026].